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# Famous Astronomers: Robert Wilson

The Cosmic Background Discovery

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## Introduction

On a spring day in 1964, at a laboratory in the New Jersey countryside, a thirty-one-year-old radio astronomer named Robert Woodrow Wilson pointed a strange, scoop-shaped antenna at the sky and found a problem. The antenna was too noisy. Not broken noisy, not fixable noisy: just a faint, unexplained hiss of radio energy, about three and a half degrees' worth of temperature, that came equally from every direction he looked, day or night, summer or winter, and refused every mundane explanation he and his partner Arno Penzias could devise. They checked their instrument. They checked their mathematics. They climbed inside the antenna, cleaned out the pigeons, and cleaned up what the pigeons had left behind. The hiss remained.

What they had stumbled into was the oldest light in the universe, the cooled afterglow of the Big Bang itself, a whisper of radio static filling all of space, predicted by cosmologists sixteen years earlier and hunted by a rival team at Princeton even as the two Bell Labs men were sweeping pigeon droppings from their horn. Within a year the measurement and the explanation had been published together, the ancient debate between an evolving and a steady universe had been effectively ended, and the field of cosmology, until then a playground of speculation, had been handed its first great observational instrument. A decade and a half later, the Nobel committee called the hiss's discoverers to Stockholm.

This book tells Wilson's story whole: the Texas boyhood and the education that made him a radio astronomer, the physics of antennas and noise that made the discovery possible, the ten months of stubborn checking that made it credible, and the decades after, in which Wilson, a meticulous experimenter to the end, helped open a second new field, the study of the cold molecular clouds where stars are born. It also tells the story around the story, because the discovery of the cosmic microwave background is one of science's great serendipities: an antenna built for telephone satellites, a pair of engineers doing calibration work nobody else wanted, a Princeton group that had the theory ready and lost the race by months, and a scientific world that took years to absorb what the hiss meant.

The reader should know at the start what kind of man occupies these pages. Wilson was not a flamboyant genius of the old astronomer mythology; he was, by every account including his own, a quiet, precise, careful man, an instrument-builder's instrument-builder, who trusted numbers more than narratives and who spent his career measuring things other people assumed were known. The discovery that made him famous rewarded exactly those virtues: patience, skepticism, and a refusal to stop checking until the check itself became the discovery. His life is a study in how science's biggest prizes actually get won, not by flashes of insight alone but by the discipline of looking harder at an inconvenience than anyone else thought necessary.

The chapters follow the familiar plan of this series. The first four trace the foundations: his youth and formation, the plain physics of the cosmic background he found, his working character, and the radio engineering behind the instrument. The middle chapters cover the discovery itself, in detail, and the world that received it. The later chapters follow his later career and legacy, the modern view of the discovery and of the neglected theorists who predicted it, the open questions his measurement opened, and what cosmology's future holds for the light he found. Technical concepts are introduced where the story needs them, in plain words, and no special knowledge is required to follow it, only the patience to follow a careful man through the noise.

A note on names, since this book's subject shares his with half a century of other notable men. The Robert Wilson of these pages is Robert Woodrow Wilson, the radio astronomer of Bell Labs; he is not the novelist, the dramatist, the stage director, the tenor, the general, or any of the others who fill the name's long page in the encyclopedias. The book's earlier edition, the reader may be forgiven for suspecting, spent its pages on that very list; this one spends them on the physicist. Where the story touches another Wilson or another discoverer, the full name and the role are given, and the reader can keep the cast straight with one rule: the Texan with the antenna is the subject, and everyone else enters only to shine light on him.

The series this book belongs to, the lives of the great astronomers, has room for many kinds of subject: the imperial mathematicians of the telescope's first age, the sky-sweeping cataloguers of the eighteenth and nineteenth centuries, the theorists who imagined what observers could not yet see. Wilson belongs to a type the series reaches only in its modern chapters: the industrial scientist, the experimenter whose laboratory was a corporation's, whose instrument was built for commerce, and whose salary was signed, indirectly, by the telephone bill. His life proves that the type belongs in the gallery, for the discovery that crowned the century of astronomy's greatest era was made neither at a mountaintop observatory nor at a great university, but on a corporate campus in New Jersey, by an employee, with the company's equipment, on the company's time. The gallery's newest wing,

in other words, is the one his story opens, and the reader who follows him through it will learn how the modern age does its astronomy.

## Chapter 1: The Discovery of Robert Wilson

Robert Woodrow Wilson was born in Houston, Texas, on the tenth of January, 1936, in an America entering the Depression's later years, and his childhood had the two ingredients that recur in the biographies of experimentalists: a family comfortable with technical things and a boyhood sky worth watching.

His father was a chemical engineer in the oil business, the industry that gave Texas its wealth and its chemistry, and the household's furniture included the tools and talk of measurement; the son absorbed early that the world was a solvable machine. The Houston of his boyhood lay at the flat, humid edge of the Gulf, its skies swept clean of northern haze, and the boy who grew up under them with a curiosity about how things worked found his way, in time, to the question of what the sky was made of. Family and church gave the era's standard moral furniture, and Wilson kept all his life the personal modesty his Texas Methodist upbringing installed; colleagues at his most famous hour would describe a man visibly uncomfortable with applause.

His formal formation was strictly first-rate. He studied physics at Rice University in Houston, then the region's great school of the sciences, graduating in 1957, and went on to California for doctoral study at the California Institute of Technology, taking his doctorate in 1962. His research there, conducted at the observatory the institute ran in the high desert mountains of eastern California, was radio astronomy, the young science that had grown since the war out of the collision of radar technology with the sky: giant dish antennas, receivers cooled and tuned, and a map of the universe in wavelengths the eye had never seen. The school of radio astronomy in that era was the best possible apprenticeship for what was coming, because it taught a discipline the optical astronomers of the day rarely needed: the absolute calibration of an instrument, the art of knowing, truly knowing, how much energy the antenna is receiving, in watts, from every candidate source, natural and unnatural, including the instrument itself.

This last art deserves a paragraph at the outset, since it is the hinge of the whole story. A radio telescope does not see; it absorbs, and converts absorbed radio power into a signal whose strength must be interpreted against everything else that heats the receiver, the ground, the electronics, the sky itself, the whole thermal mishmash that radio engineers lump together as noise and measure, perversely but conveniently, as an equivalent temperature. A young astronomer trained to distrust every kelvin of that noise, to account for it, subtract it, and measure the remainder, is a young astronomer trained exactly to notice when the arithmetic refuses to close. Wilson's doctoral apprenticeship, spent measuring and mapping radio sources with careful attention to their true brightnesses, was, in retrospect, a years-long rehearsal for the ten months in 1964 and 1965 when the noise would not close, and the remainder would be the universe.

His road from the desert to New Jersey ran through industry's golden doorway. In 1963 Wilson joined the research arm of the Bell Telephone Laboratories, working at the laboratory's rural campus at Crawford Hill near Holmdel, New Jersey, the same hills where, a generation earlier, Karl Jansky had accidentally founded radio astronomy itself while chasing static for the telephone company. Wilson went to Bell Labs for the tools: the company that operated the nation's long-distance radio links had built, for its satellite experiments, some of the finest low-noise antennas on Earth, and its laboratory had a standing tradition, generous and unusual, of letting scientists use the equipment for the sky. He was assigned to work with a slightly older colleague, a German-born physicist named Arno Penzias, on using the lab's remarkable antenna for radio astronomy. The two men, both in their early thirties, both product-trained in absolute measurement, settled in at Crawford Hill, and began, in the spring of 1964, to measure the sky with the best instrument either had ever touched.

The instrument was a horn antenna, twenty feet across its mouth, shaped like an enormous sugar scoop turned skyward, and its history explains its presence. It had been built years earlier, by the laboratory's engineers, for the Echo project, the experiment in which the telephone company bounced radio signals off a giant metal balloon orbiting overhead, a first demonstration that space itself could carry a long-distance call; its designers, seeking every fraction of a decibel, had shaped and shielded it to be astonishingly quiet, sensitive, and free of the ground's interfering warmth. The satellite projects had moved on, the balloon era had passed its experimental peak, and the horn sat at Crawford Hill like a retired racehorse, ready for anyone careful enough to ride it. Wilson and Penzias intended to use it for a modest program: an accurate measurement of the sky's radio brightness, in several directions, in the centimeter band, calibrated against a standard reference of their own devising. It was to be the first really trustworthy measurement of the high galactic latitudes, the parts of the sky away from the Milky Way's bright plane, where the radio sky was thought to be almost empty.

They began in May 1964. Within weeks, they had their problem: the sky measured too warm. And so begins the second discovery of Robert Wilson, the one the world remembers.

The Caltech years deserve one more memory, because they fixed the young Texan's scientific taste. The observatory in the eastern Sierra, its dishes strung across a high desert valley, was in that era the world's radio school: its staff, drawn from the great war-generation experimenters, ran the place with a culture of absolute measurement and instrument-building that a doctoral student absorbed like weather. Wilson's own project, the measurement of radio sources' true brightnesses, put him inside that culture at its most exacting, and his memoirs credit the years with the habits that would matter at Holmdel: the distrust of borrowed calibrations, the habit of building one's own standards, and the standing question, applied to every number, of what exactly was being compared with what. The schools of science are named for their graduates' habits, and the reader who wants to know which school made the background's discoverer may watch him, in every later description of his work, asking the desert's old questions about every antenna he touched.

The Houston boyhood deserves one more detail, the one biographical fact the man himself has offered most often: the boy watched the sky. The Texas nights, the flat horizon, and a household that rewarded questions produced the standard astronomer's origin, a child staring upward and wondering, and the specific object the memoirs mention is the Moon, the standard first love. What the detail should teach the reader is not the psychology, which the biographer can only guess at, but the continuity: the man who measured the universe's oldest light began as the boy who could not stop looking up, and the sixty years between were spent in a single widening of that first activity, from the Moon to the stars to the static between them. The series' subjects, taken together, show the same curve, generation after generation, the eye that would not close; Wilson's chapter of it is simply the most recent, and the reader who finishes this book and stares upward tonight is repeating, in the first motion, the whole lineage's first motion.

The move to Bell Labs deserves one more beat, the employer's own history with the sky, since it explains why a telephone company had the instrument at all. The laboratory's radio astronomy was a matter of inheritance: static, the enemy of communications, had been studied at the company since the 1930s, when Karl Jansky's systematic hunt for the hiss that troubled transatlantic telephone calls found the galaxy emitting at radio wavelengths and founded a science by accident; the research arm, its attention turned to the sky by the needs of the wire, had kept a radio-astronomy culture alive for three decades, funding antennas and hiring scientists for reasons that mixed commerce with curiosity. By the time Wilson arrived, the tradition was ripe: the company's satellite experiments had generated instruments beyond the universities' means, and the research culture permitted their scientific use. The young Texan who walked onto the Crawford Hill campus in 1963 was thus entering a lineage as well as a laboratory, Jansky's heir with better tools, and the reader who follows the lineage can feel the symmetry: the static that founded the science in one decade was explained by the static that crowned it in another, and both discoveries were made, in the plainest sense, on the telephone company's clock.